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Optimality-theoretic advances in our understanding of Spanish syllable structure*

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This article offers an optimality-theoretic account of Spanish syllable structure that integrates all areas covered by traditional analyses of syllabification: syllable types, phonotactic restrictions, syllabification algorithms, domain of syllabification, and repair strategies. The article aims to highlight the advantages of an optimality theoretic approach to syllabification in order to support the claim that OT has brought about significant improvement in syllabic theory. Among these advantages are the replacement of language-specific rules with universal constraints; the resolution of rule conspiracies; and the elimination of stipulatory statements (quality of epenthetic vowels) and adhoc conditions on rule application. The descriptive facts are explained through the general mechanism of interaction of universal markedness and faithfulness constraints. Cross-dialectal and intra-speaker variation are obtained through variation in the ranking of the constraints. By resorting to universal constraints and language-specific ranking, variation is shown to result from the implementation of various ways of obtaining the same goal (e.g., elimination of coda consonants: complete deletion, voice neutralization, stricture neutralization, vocalization, etc.). Processes that were presented as separate in a derivational model (i.e., Spanish diphthongization, resyllabification and onset strengthening) are shown to respond the same motivation (avoid onsetless syllables). The article aims to be an updated optimality-theoretic alternative to general accounts of syllabification such as Harris (1989) and Hualde (1991).

Keywords: syllabification, diphthong, hiatus, syllable, syllable structure, Optimality Theory, phonotactics

o. Introduction. The role of the syllable in phonological theory

Since its introduction as a unit of phonological organization in the late seventies (Hooper 1976, Kahn 1976), the syllable has played an increasingly important role in

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phonological theory. The syllable allows the analyst to formulate generalizations that could not be easily expressed without it. For instance, consider some well-known processes of Spanish phonology, such as nasal velarization (e.g. [p̃aŋ] ← /pan/ 'bread'), /s/ aspiration (e.g. [doh] ← /dos/ 'two'), /s/ voicing assimilation (e.g. [mizmo] ← /mismo/ 'same'), liquid gliding (e.g. /papel/ [papeḷ] ← /papel/ 'paper'), and nasal assimilation (e.g. [kambio] ← /kanbio/ 'change'). Without referring to the syllable, the analyst would fail to grasp the common underlying motivation for those processes: the neutralization of contrasts in a weak syllabic position (coda). Similarly, phenomena like onset strengthening, in which a high vocoid is realized as a palatal obstruent in onset position (e.g., /kom-iendo/ [ko.mjeṇ.do] 'eat-gerund' vs. /kre-iendo/ [kre.yeṇ.do] 'believe-gerund') can be seen as the result of the onset preference for strong, less sonorous segments, an insight that would be entirely lost in a model of phonology without syllables.

The syllable is also relevant with respect to non-segmental aspects of the phonology. For instance, many stress generalizations cannot be formulated without it. In Spanish it is well-known that lexical stress cannot be placed further to the left than the antepenultimate syllable (3-syllable window, Harris 1983 and many others after him), *teléfono* **teléfono* 'telephone', or further than the penultimate when the latter has a branching rhyme (branching condition, Harris 1983), *Venezuela* **Venezuela*. Phonotactic restrictions also hold over syllables and syllabic components with regard to the type and number of segments. For example, the illformedness of hypothetical **muersto* can only be explained as the result of an excessive number of segments in the rhyme (three being the maximum), given that /ue/ and /rs/ sequences are by themselves licit strings.¹

The importance of the role played by the syllable in modern phonological theory increased exponentially with the advent of Optimality Theory (OT). The syllable is possibly the area of phonological inquiry that has attracted the most interest amongst OT-practitioners and, as a result, it constitutes the focus of a significant body of OT research. All of the early OT studies contain large sections devoted to syllabification issues (cf. Prince and Smolensky 1993, McCarthy and Prince 1993a & b). In Spanish too the first dissertations published within an OT framework were studies of syllabification and syllabification-related processes (Morales Front 1994, Colina 1995). The volume of syllable-related OT research does not come as a surprise if one considers the ease with which syllable types illustrate optimality accounts of factorial typologies and how the syllable serves to exemplify the interaction of segments, moras, sonority, edges, and stress in OT.

In addition to contributing to the development of OT, the syllable has benefited from advances in OT, in particular, OT has helped to improve our understanding of the syllable and syllable structure. In general terms, three areas of the theory have proven useful for syllabic theory:

(a) In OT language-specific rules are replaced by universal constraints and language-specific constraint ranking. Consequently, *syllable structure is the result of*

1. *St-* is not an acceptable onset cluster in Spanish.

conflicting universal requirements on syllable structure prioritized differently across languages. An important consequence of this is that OT can formalize and account for the similarity of syllabification mechanisms in various languages in a much more principled way than the numerous language-specific syllabification rules of serial phonology. For example, most languages seem to have an *Onset Rule* that attaches one or more consonantal segments to the left of the nucleus. Yet, in order to capture the universal nature of this rule, a derivational model, consisting of language-specific rules, has to stipulate that the *Onset Rule* is universal. In OT, however, the universal preference for V.CV is the effect of a universal constraint, ONSET, that requires that all syllables have an onset, in combination with a universal dislike/constraint for codas, *CODA. Another important contribution of OT to the understanding of syllable structure is that constraints are violable under domination from a more highly ranked constraint. Hence not all languages and all syllables will have onsets — some languages will have onsetless syllables. The prioritizing of conflicting, violable, universal constraints in multiple rankings produces variation in syllable structure and syllabification.

(b) In OT constraints are of two types: those that require preservation of underlying contrasts (faithfulness constraints) and those that favor unmarked structure. Thus, syllable structure (i.e., the output of syllabification/a syllabified output) is *the result of the interaction of universal faithfulness and markedness constraints*. ONSET, for example, is a markedness constraint that states that the preferred syllable type is CV. If all syllable markedness constraints were always dominant across languages, languages would exhibit no marked syllable structure and the only syllable type would be CV. However, this is not so, because in order to be useful communication tools, languages need to express a large number of contrasts. In response to this functional need, faithfulness constraints require preservation of input form. Hence, inputs may consist of syllables without onsets. If the corresponding faithfulness constraint (DEP-IO — no epenthesis — or MAX-IO-no deletion) is ranked above markedness, the output and the language will permit onsetless syllables. If markedness is ranked above faithfulness (DEP-IO or MAX-IO), all syllables will have onsets. An important observation brought to light by OT is that when the faithfulness constraint(s) forcing violation of markedness is/are no longer relevant, unmarked structures will surface (“the emergence of the unmarked”, McCarthy and Prince 1994). This has been shown to be true in adult phonology as well as in child and second language phonology. In Spanish hypocoristics, for instance, faithfulness to the underlying representation is no longer at stake because hypocoristic formation implies deletion by definition. CVC or CG(glide)V syllables, usually tolerated because of the domination of faithfulness over markedness constraints (*CODA, *COMPLEX-NUCLEUS), are truncated to CV when faithfulness is no longer dominant, e.g. *Constantino* > *Consta*, [daniɛl] > *Dáni* ‘Daniel’ (Colina 1996, Piñeros 2000).

(c) OT can satisfactorily explain *conspiracies*. Languages often have several rules that work towards a common goal. For instance, Spanish has a rule of diphthongization (i.e., syllable merger) of high vowels when preceded or followed by another vowel, *mi amigo* [mja.mi.yo] ‘my friend’.

$$(1) \begin{pmatrix} -\text{cons} \\ +\text{high} \end{pmatrix} \rightarrow [-\text{syllabic}] / \left\{ \begin{array}{l} \text{V} (\#) ______ \\ ______ (\#) \text{V} \end{array} \right\}$$

It also has a rule that resyllabifies a word-final consonant to the onset position of a vowel initial word, *tus amigos* [tu.sa.mi.yos] ‘your friends.’

$$(2) \begin{array}{ccc} +\text{cons} & \rightarrow & +\text{cons} / ______ \# \text{V} \\ | & & | \\ \text{coda} & & \text{onset} \end{array}$$

(Harris 1983: 43–44)

Although not apparent from the format of the rules, (1) and (2) have as a common goal the elimination of onsetless syllables. Serial rules and derivational formalisms have been known to hide the real purpose behind rules, in what is usually referred to as “the conspiracy of rules” (Kenstowicz and Kisseberth 1979). OT, on the other hand, naturally and elegantly captures this “purpose” by means of interacting universal constraints. In the examples in (1,2), it is the domination of *ONSET* over other relevant constraints (e.g. *ALIGN-L*, *MAX-IO_μ*) that drives both resyllabification and syllable merger in the outputs: *ONSET* satisfaction is obtained by misaligning word and syllable boundaries (*ALIGN-L* violation) and by underparsing a mora (*MAX-IO_μ* violation), respectively. Our understanding of syllabic structure has improved as the result of OT’s ability to bring forth the shared motivation behind many rules of syllabification.

In sum, while syllable structure has been crucial in the development of modern phonology and in particular of OT, OT has in turn and, perhaps even to a greater extent, contributed to advancing our understanding of syllable structure. Within this context, the purpose of the current chapter is to review and highlight recent developments and advances brought about by OT with regard to the syllable structure of Spanish. The goal is not to compare fine details of analyses of Spanish syllable structure, but to show in general terms the greater explanatory power of OT with respect to Spanish syllabification. Hence many matters of detail will not be covered. A second goal of this paper is to offer a comprehensive overview of syllable structure in Spanish that is inclusive of the fifteen years of work since the publication of similar studies (Harris 1989a, Hualde 1991) and that examines optimality theoretic progress in this realm. This chapter also provides a general overview of syllable types, syllabification algorithms and other syllable-related phonological phenomena within an OT framework.

The present article is organized as follows: the section just concluded is an introduction dealing with the role of the syllable in phonological theory and more specifically, its role within OT. Using Spanish data as examples, the introduction highlights improvements in our understanding of syllable structure derived directly from OT. Section 1 offers an OT account of general syllable types and phonotactic generalizations derived from the parsing of segments into syllabic constituents. For purposes of comparison, Section 2 presents a review of syllabification rules in a serial model. It also provides an optimality theoretic model of resyllabification. Finally, Sections 3 and 4 complete the picture by examining the interaction with morphology (in serial models, domain of syllabification) and syllable repair mechanisms. The chapter concludes

with some conclusions and a general overview of serial vs. constraint-based accounts of syllable structure.

1. Syllable types and phonotactics in Spanish

1.1 Basic syllable types

In purely descriptive terms, most dialects of Spanish have the syllable types in (3):

(3) Syllable types in Spanish (in italics in the examples)

V	<i>a</i> .la	'wing'
CV	co. <i>lor</i>	'color'
CVC	pa <i>n</i>	'bread'
VC	un	'one'
CCV	fl <i>o</i> .tar	'float'
CCVC	tr <i>en</i>	'train'
VCC	ins.tru <i>ir</i>	'instruct'
CVCC	pers.pec. <i>tiva</i>	'perspective'
CCVCC	trans.por. <i>te</i>	'transportation'

For purposes of presentation, I do not include glides (G) in the typology at this time.

In OT, the possible syllable types of Spanish are the result of the interaction of universal violable constraints (markedness and faithfulness). The universal nature of such constraints makes the account superior to the purely descriptive list given in (3). That these syllable types constitute a limited inventory from which all languages draw their syllable templates finds an explanation in the existence of a limited number of constraints present in all languages. Given these assumptions about the universal nature of constraints, the analyst must demonstrate that the relevant constraints are truly universal (see Kager 1999 for detailed justification). Briefly, a universal of syllable typology is that all languages prefer syllables that start with a consonant (onset) and end in a vowel, CV being thus the preferred, unmarked syllable type. While some languages allow codas, none require them; similarly, some languages allow onsetless syllables, but none forbid onsets. In sum, there are no languages that require codas and forbid onsets. Thus, OT argues for two universal markedness constraints:

(4) ONSET: No vowel-initial syllables

*CODA: Syllables cannot end in a coda²

2. Itô and Mester (1994) propose reformulating syllable well-formedness constraints as alignment conditions ALIGN-L(σ , C) and ALIGN-R(σ , V). Piñeros (2001) does this for Spanish. The interested reader is referred to Piñeros (2001) and Piñeros (this volume) for details. Generally, the Onset and Coda markedness hierarchies in (21) and (22) correspond to the alignment hierarchy ALIGN-L(Obstruent, σ) >> ALIGN-L(Nasal, σ) >> ALIGN-L(Liquid, σ) >> ALIGN-L(Glide, σ) in Piñeros (2001). A comparison of the two formalisms is beyond the scope of this paper. The non-alignment format was chosen here for its greater presentational simplicity and for its wider use within OT.

In addition, complex syllabic constituents are universally more marked than singletons: there is no language that has complex onsets/codas/nuclei and lacks single segment onsets, codas and nuclei. The reverse, however, is true. There exist languages that ban complex subsyllabic components but allow singletons. The following universal constraints are therefore proposed by OT:

- (5) *COMPLEX ONSET: No more than one segment in the onset
 *COMPLEX CODA: No more than one segment in the coda
 *COMPLEX NUCLEUS: No more than one segment in the nucleus

Spanish allows onsetless syllables as well as syllable codas

- (6) Basic segmental faithfulness constraints
 MAX-IO: Every segment present in the input must have a correspondent in the output (anti deletion constraint).
 DEP-IO: Every segment present in the output must have a correspondent in the input (anti epenthesis constraint).
 FAITH: (cover term for all faithfulness constraints)

Therefore, ONSET and *CODA must be dominated by faithfulness constraints: FAITH >> ONSET, *CODA.

- (7) /ala/ [a.la]

	FAITH	ONSET	*CODA
a. $\varnothing$ a.la		*	
b. Ta.la	*!		
c. ala		**!	*
d. la	*!		

- (8) /poner/ [poner]

	FAITH	ONSET	*CODA
a. $\varnothing$ po.ner			*
b. po.ne.re	*!		
c. pon.er		*!	*
d. po.ne	*!		

In (7) candidates with epenthesis and deletion violate the top-ranked FAITH because they alter the input representation either by inserting an element not present in the input (epenthesis of T, [7b]) or deleting an input phone (7d). (7a) and (7c) incur ONSET violations on account of one and two onsetless syllables, respectively. By syllabifying the intervocalic consonant with the second vowel, candidate (7a) incurs only one ONSET violation, while at the same time avoids a contravention of *CODA and is thus more optimal than (7c). Therefore, although not perfect, (7a), with only one ONSET violation, is the optimal candidate and the output.

Evaluation proceeds similarly in (8). [poner] (8a) is the optimal candidate because (8b) [ponere], an attempt to get rid of the coda consonant through epenthesis, violates the more-highly ranked FAITH. (8d) encounters the same fate given that there is deletion of the final consonant (a consonant present in the input is absent in the

output). (8c) incurs an unnecessary violation of ONSET and is therefore eliminated. It could be argued that word-initial onsetless syllables (*a.la*) are not the result of faithfulness, but the effect of an alignment constraint that bans insertion in word-initial position (McCarthy and Prince 1993 a&b; Kager 1999: 111).

- (9) ALIGN-L: The left edge of the grammatical word coincides with the left edge of the prosodic word

However, the existence of word-internal onsetless syllables serves to discard this hypothetical ranking, *ma.re.a* 'tide', *ve.o* 'I see'.

As mentioned above, Spanish also allows complex onset and codas, which suggests that FAITH dominates the markedness constraints prohibiting onset and coda clusters *COMPLEX ONSET, *COMPLEX CODA.

- (10) FAITH >> *COMPLEX ONSET, *COMPLEX CODA.

- (11) /broma/ [bro.ma] 'joke'

	FAITH	*COMPLEX ONSET
a. ^σ bro.ma		*
b. ro.ma	*!	
c. be.ro.ma	*!	

- (12) /insto/ [ins.to] 'I urge'

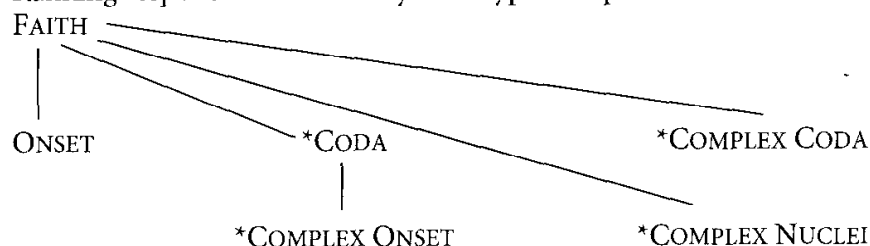
	FAITH	*COMPLEX CODA
a. ^σ ins.to		*
b. in.to, is.to	*!	
c. is.to	*!	
d. i.nes.to	*!	
e. in.se.to	*!	

*CODA in turn must dominate *COMPLEX ONSET given that word-internal onset clusters are preferred to the parsing of the consonants in the coda and the onset of two adjacent syllables (13) (cf. *onset maximization* in derivational approaches). *COMPLEX NUCLEI is violated by diphthongs, /pierce/ → [piérðe]. I return to them in 1.2.1.

- (13) /potro/ [po.tro] 'young horse'

	*CODA	*COMPLEX ONSET
a. ^σ po.tro		*
b. pot.ro	*!	

- (14) Ranking responsible for basic syllable types in Spanish



1.2 Segments and the syllable

Despite its general explanatory adequacy, careful consideration reveals that the general ranking in (14) is not sufficient to account for all the facts. Not all consonant clusters in VCCV sequences are parsed in the same syllable, *par.te* * *pa.rte* 'part'; similarly, although Spanish allows consonants in the coda, not all codas are acceptable in all dialects, *club* [*klub] 'club'. It is apparent that syllabic constraints that refer to subsyllabic components and number of segments are not sufficient. Minimally, generalizations about the specific segments parsed in each of those positions must be accounted for as well as the sequence (order) of segments within the syllable. In broad terms, two areas need to be addressed: sonority (1.2.1) and the parsing of segment types in subsyllabic constituents (1.2.2).

1.2.1 *The sonority contour*

Syllables exhibit a contour based on the universal sonority scale in (15) according to which the syllable rises in sonority towards the nucleus (the most sonorous point) and then decreases towards the coda (Sonority Sequencing Principle, Clements 1990). Segments are ranked according to sonority in the universal sonority scale, which coincides with the one proposed for Spanish in Harris (1989b):

- (15) Obstruents < Nasal < Liquid < Glide < Vowel

For Spanish, Martínez-Gil (1996, 1997) has proposed a subset of distinctions that separates obstruents into stops and fricatives.

- (16) Stops < Fricatives < Nasal < Liquid < Glide < Vowel

In most dialects of Spanish, stops, fricatives, and nasals are possible onsets, but not glides or vowels. Only vowels are possible nuclei (glides are well-formed in prevocalic position). All segment types can be parsed in the coda except for vowels.

Pre-OT models of phonology express these facts in the form of descriptive generalizations, language-specific rules ('mark vowels as syllable heads') or restrictions on the application of the rule (Harris 1983, 1989a; Hualde 1991, Hualde 1999a). Additional rules are required to express repair mechanisms. Sonority could not be formalized in a direct way without resorting to special conditions stipulating that a rule would apply only if certain sonority conditions were met (for proposals and discussion on the formalization of sonority within a derivational model, see Selkirk 1984, Clements 1990, Rice 1992). On the other hand, an optimality-theoretic framework captures the generalizations that relate syllabic positions with sonority classes in a straightforward manner by means of universal scales and constraint hierarchies. The syllable-position prominence scale (Nucleus > Coda > Onset >) can be easily combined with the sonority scale (via harmonic alignment) to produce a constraint hierarchy (McCarthy 2002: 21). The most prominent syllable position (Nucleus) is aligned with the most sonorous segment (Vowel) and viceversa, so that the most highly ranked constraint is the one that penalizes parsing the most sonorous segment in the least prominent position.

- (17) Harmonic Alignment of Prominence Scales (sonority and syllabic prominence)
- | | |
|-----------------|-------------|
| Nucleus > | > Onset |
| | |
| Vowel > | > Obstruent |
- (18) Constraint Hierarchy
- *NUC/obstruent >> >> *NUC/vowel
- *ONSET/vowel >> >> *ONSET/obstruent

The ranking of faithfulness constraints with regard to the sonority scale (hierarchy of markedness constraints) determines whether violations can be forced through domination and explains language variation (language-specific ranking) in an otherwise universally fixed hierarchy. Thus, in Spanish, the fact that only vowels can be nuclei is the result of the ranking in (19). Since the constraints against parsing non-vocalic segments in the nucleus are undominated, only vocalic nuclei are possible.

- (19) Segment types in the nucleus
- *NUC/obstruent >> *NUC/nasal >> *NUC/liquid >> FAITH, *NUC/vowel

This is not the case in languages, such as English, that allow syllabic liquids and nasals. In English a liquid or nasal can serve as a syllable nucleus if no vowel is available. The domination of FAITH over *NUC/Nasal >> *NUC/liquid guarantees the parsing of a nasal or liquid in the nucleus rather than deletion or vowel epenthesis.

- (20) *NUC/obstruent, FAITH >> *NUC/nasal >> *NUC/liquid >> *NUC/vowel

With respect to the onset position, Spanish sonorants and obstruents can appear in the onset, but obstruents are preferred.³ Vowels cannot be in the onset and therefore *ONSET/vowel must be undominated. This is reflected in the ranking in (21).

- (21) Segment types in the onset
- *ONSET/vowel >> FAITH >> *ONSET/liquid >> *ONSET/nasal >> *ONSET/obstruent

In the coda, as sonority decreases from the nucleus, the more sonorous sonorants are preferred to the obstruents as shown in (22). Vowels are not possible codas as indicated by the undominated status of *CODA /vowel.

- (22) Segment types in the coda
- *CODA/vowel >> FAITH >> *CODA/obstruent >> *CODA/nasal >> *CODA/liquid >> *CODA /glide

Glides have been purposefully omitted from the sonority scales in (19–21). In Spanish prevocalic glides are generally part of the nucleus and postvocalic glides are parsed in the coda (see Harris 1983, Hualde 1991 and Núñez-Cedeño and Morales Front [1999: 174–175] for arguments). In addition to *NUC/glide, prevocalic glides incur a *COMPLEX NUCLEUS violation (24a). This is done in order to avoid violating the more-highly ranked ONSET (24b & d).

3. Evidence for this can be found in child language acquisition. In early acquisition children often avoid liquid and nasal onsets (Barlow and Gierut 1999, Ohala 1999, Gnanadesikan 2004).

(23) *ONSET/glide, ONSET >> *COMPLEX NUCLEUS, *NUC/glide

(24) /pierce/[piérðe] 'he loses'

	*ONSET/glide	ONSET	*COMPLEX NUCLEUS	*NUC/glide
a. $\text{pi}^{\text{e}}\text{r}.\text{ðe}$			*	*
b. $\text{pi}.\text{ér}.\text{ðe}$		*!		
c. $\text{pj}^{\text{e}}\text{r}.\text{ðe}$	*!			
d. $\text{pj}.\text{ér}.\text{ðe}$	*!			

That the glide is not parsed in the onset is guaranteed by the undominated constraint *ONSET/glide (24c).⁴ [j] is used in (24c) and in the rest of this paper to represent an onset glide, thus graphically distinguishing it from a nuclear glide, as in (24a).

The careful reader will have noticed that the tableaux in (24) is not quite complete. In order for an underlying vowel to be parsed as part of a complex nucleus and also satisfy ONSET, its input mora must not be present in the output, thus incurring a violation of FAITH, more specifically MAX-IO μ (a mora present in the input must have a correspondent in the output).

(25) Diphthongization

*ONSET/glide, ONSET >> *COMPLEX NUCLEUS, *NUC/glide, MAX-IO μ

Glides cannot be the sole segment in the nucleus. However, no additional constraint is necessary at this point, since, in order to avoid a violation of the highly-ranked ONSET, glides are parsed in a complex nucleus, thus effectively avoiding sole nuclear membership, as seen in (24a) vs (24d).^{5,6}

Given the undominated status of *ONSET/glide, glides can never be parsed in the onset. However, an underlying vocoid can be forced into the onset (thus satisfying ONSET) through faithfulness violations — MAX-IO μ and IDENT (son), e.g., /kom-iendo/ [ko.mjén.do] 'eat-gerund' vs. /kre-iendo/ [kre.yen.do] 'believe-gerund'.

(26) IDENT (son): The specification for the feature [sonorant] present in the input must match the one in the output.

4. For falling diphthongs, *peine* [pej.ne] 'comb', evaluation proceeds in the same fashion, except for the replacement of *NUC/glide (trivially satisfied) with *CODA/glide, *CODA.

5. Despite this, an argument could be made for an undominated constraint requiring all nuclei to be moraic. (24d) would violate this constraint.

6. An issue not addressed in this paper is that of the reported contrast between diphthongs and hiatuses in some varieties of Spanish, *Luí.sa* vs. *Su.í.za* 'Switzerland' (Hualde 1997, Hualde 1999b, Colina 1999, Hualde and Prieto 2002). In a limited number of examples in Spanish, a hiatus is found instead of the expected diphthong, thus giving rise to a few near minimal contrasts like the one above. Many of these forms can be explained by means of analogy (IDENT constraints, Colina 1999), but a few remain unexplained. Given the very limited number of these contrasts in the phonology, I adhere to the position that they are exceptional and that the diphthongs hiatus distinction is generally predictable (Hualde 1997, Hualde and Prieto 2002).

(27) Onset strengthening

*ONSET/glide, ONSET >> MAX-IO_μ, IDENT (son)

(28) /kre-iendo/ [kre.yeɲ.do]

	*ONSET/glide	ONSET	MAX-IO _μ	IDENT (son)
a. kre.yeɲ.do			*	*
b. kre.iéɲ.do		*!	*	
c. kre.jéɲ.do	*!		*	
d. kre.i.eɲ.do		**!		

(28) shows that the process traditionally known as ‘onset strengthening’ is simply the result of trying to satisfy ONSET while also avoiding violations of the high-ranked markedness constraint against glides in the onset (*ONSET/glide). Faithfulness violations are preferred in this case. Parsing the glide in the onset (28c) or in the nucleus (28b) incurs violations of top-ranked *ONSET/glide and ONSET. A candidate like (28d), in which /i/ is a vowel, is ruled out because of two ONSET violations. The fricative in (28a) is a better option since, by changing its stricture and underparsing a mora, (28a) only violates the low-ranked faithfulness constraints IDENT (son) and MAX-IO_μ. An OT analysis of this kind solves the problems of causality faced by serial models in which it is not possible to ascertain whether a glide becomes an obstruent after it has moved into the onset position, or whether the glide moves to the onset after it has become an obstruent and therefore is no longer acceptable in the nucleus (Hualde 1991). Similarly, a derivational framework offers at least two equally plausible analyses of onset strengthening: (a) the high vowel is initially parsed as a glide, and then it moves from the nucleus to the onset by means of rule that modifies the relevant autosegmental associations; and b) alternatively, the high vowel is parsed as part of the nucleus if it is postconsonantal, but as part of the onset if it is (absolute) syllable-initial, thereby feeding glide strengthening. In contrast, OT offers one single, more straightforward and less stipulative account of the phenomenon, as shown in (28).

As seen above (24–25, 27–28), moraic structure interacts with syllabic constituency and the sonority hierarchy ranking. Nuclei are always moraic, onsets always non-moraic and codas can be moraic or non-moraic. Faithfulness violations with regard to moraic structure (DEP-IO_μ and MAX-IO_μ — insertion or deletion of moras) can be incurred under pressure from more highly ranked constraints (e.g., ONSET). In Spanish, high vocoids become fricatives and complex nuclei and codas (vowel + glide, glide + vowel) are allowed to avoid onset violations (e.g. diphthongization, *diario* [d̪i̯a.ɾio] ‘diary’; onset strengthening, [28]). A complete account of these processes, however, also needs to explain the selection of the vocoid that loses its mora (i.e., incurs a MAX-IO_μ violation). While derivational models simply referred to the degree of sonority (‘the least sonorous vocoid is the one that becomes a glide’ or, in other words, those vocoids that are the least likely to be associated with a mora), an OT analysis can elegantly formalize this generalization by means of a constraint hierarchy based on the universal sonority scale.

- (29) *ONSET/glide, ONSET >> *COMPLEX NUCLEUS, *MAX-IO_μ
 *ONSET/glide, ONSET >> *IDENT (son), *MAX-IO_μ
 *hi/μ>> *mid vowel/μ>> *low/μ

The high vowels are the ones that are more likely to lose their mora since their moraicity incurs a higher violation than mid vowels and low vowels; mid glides are better than low glides because their moraicity incurs a higher violation than low vowels.

Before concluding this section, a few words must be said about the sonority of non-nuclear clusters. Onset and coda clusters must obey the sonority sequencing principle (SSP) and therefore onset clusters that do not increase in sonority or coda clusters that do not decrease in sonority are not allowed; in other words, SSP is undominated, **part.e*, **pa.rte* ‘part’. In addition, the two segments of an onset cluster cannot be too close in sonority, **fu.tbol* ‘soccer’: one of the members must be an obstruent or /f/ and the other a liquid.⁷ In pre-OT models, this generalization was expressed as a condition on the application of a Complex Onset rule that attached a second consonant to the left of an already existing onset. A number reflecting the distance on the sonority scale between the two members of the cluster (stops and liquids) was used to capture the sonority distance required. In OT the onset cluster generalization is captured by a sonority distance constraint (Maximum Sonority Distance [MSD]) that requires that the first member of the cluster be drawn from the set of the least sonorous permissible onsets and the second from that of the most sonorous permissible onsets, thus maximizing the sonority distance between the two; in other words, the MSD constraint requires that there exist the maximum sonority distance between possible onset segments. Which segments are possible onsets is captured by the independently needed sonority constraint hierarchy in (21) repeated here as (30).

- (30) *ONSET/vowel, *ONSET/glide >> FAITH >> *ONSET/liquid >> *ONSET/nasal >>
 *ONSET/obstruent

The class of obstruents needs to be divided in two sonority classes, that of stops and /f/, and that of fricatives; the common property between stops and /f/ that allows them to belong to the same sonority class is that they are “the class of obstruents that lack the specification [+continuant] (Martínez-Gil 2001: 217).

MSD is undominated.⁸ It in turn dominates the faithfulness constraint against epenthesis — DEP-IO — (33), given that [eslaβo] is preferred to [slaβo] as the output of /slabo/. More specifically, the ranking MSD >> DEP-IO accounts for the well-known cases of initial epenthesis, *Yugoslavo* ~ *eslavo* ‘Yugoslavian’ ~ ‘Slavian’.

7. In addition, /dl/, and /tl/ in some dialects, are not possible onset clusters. These segments incur OCP violations regarding the sharing of voiced and place features.

8. In fact, MSD is dominated. The constraint(s) responsible for the fricative must dominate MSD in at least one possible analysis. Although the non-stop allophones of voiced obstruents have traditionally been considered fricatives (Hualde 1991), recent phonetic and phonological evidence points at their approximant nature. In this paper, I have adopted the more traditional fricative view for reasons of consistency with the research referred to. Replacing [β, ð, γ] with [β, ð, uɣ] has no consequences for the arguments/analysis presented here.

1.2.2 *Segments and syllabic positions*

Sonority, subsyllabic constituency and moraic structure cannot explain all the descriptive facts of Spanish syllabification. Harris (1989a:155) says that the sonority scale “must be supplemented by language-particular restrictions of some kind.” In this paper I show that there is no need for language-particular restrictions or descriptive conditions on permissible phonemes; instead, those generalizations can be captured by the language-specific ranking of conflicting universal markedness and faithfulness constraints. Since no additional constraints or rankings are necessary to account for onset and nucleus segments, other than those in 1.2.1, I focus on coda segments.

1.2.2.1 Segments in the syllable coda

1.2.2.1.1 *Word-medial codas.* Serial analyses of Spanish syllabification contain descriptive statements that list the consonants that are well-formed in the coda. Although a descriptive generalization of the data is always a good starting point, the problem with that type of analysis is that it does not move beyond the level of description. As indicated above with regard to onset clusters, derivational models of syllabification must tack these generalizations onto syllabification rules, e.g., Coda Rule: “adjoin a coda to the right of the nucleus. *Possible codas are....*” I show here that in OT these conditions are simply the result of the interaction of universal markedness and faithfulness. The list of licit coda segments can thus be obtained through constraint interaction. I start by stating the generalizations to then show how OT formalizes them.

Word-internally in coda position most Spanish dialects allow nasals with the same point of articulation as the following consonant, *cambio* ‘change’; /l/ or /r/ (neutralized in Caribbean dialects) *malta* ‘malt’, *marca* ‘mark’; /s/ *pista* ‘clue’; and /θ/ *pi[θ]ca* ‘pinch’ (in dialects with /θ/ in their inventories). Coda stops, lost at one point in the history of the language, have been recovered in some dialects, in formal registers and under the influence of spelling. In these cases, however, [voice] and [continuant] are neutralized, hence there is no difference between the coda obstruents in *concepción* ‘conception’ and *obsesión* ‘obsession’, *étnico* ‘ethnic’ and *administrar* ‘to administer’; *técnica* ‘technique’ and *dogmático* ‘dogmatic’ (Hualde 1989a).

Since Spanish has codas, *CODA must be dominated. I focus on obstruents first to later return to the sonorants. According to the sonority hierarchy (31), obstruents make the worst codas. This is in agreement with the descriptive facts for Spanish.

- (31) *CODA /obstruent >> *CODA /nasal >> *CODA /liquid >> *CODA /glide⁹

Many dialects of Spanish do not allow coda obstruents at all. Many of these dialects avoid them through deletion, indicating that the ranking must be:

- (32) *CODA /obstruent, DEP-IO >> MAX-IO

9. In an alignment format (Piñeros 2001): ALIGN-L(Obstruent,σ) >> ALIGN-L(Nasal,σ) >> ALIGN-L(Liquid,σ) >> ALIGN-L(Glide,σ).

- (33) MAX-IO: Every segment present in the input must have a correspondent in the output
 DEP-IO: Every segment present in the output must have a correspondent in the input

- (34) /obsoleto/ [osoletto]

	*CODA /obstruent	DEP-IO	MAX-IO
a. $\varnothing$ o.so.le.to			*
b. ob.so.le.to	*!		
c. o.be.so.le.to		*!	

As seen in (34), candidates that either preserve (34b) or avoid the coda obstruent through epenthesis (34c) are ruled out because they incur violations of the top-ranked constraints *CODA/obstruent and DEP-IO.

Formal varieties of Spanish (Hualde 1991) that retain coda obstruents exhibit the ranking in (35), thus selecting [ob.so.le.to] as the output (*CODA /obstruent violation) in (36):

- (35) DEP-IO, MAX-IO >> *CODA /obstruent

- (36) /obsoleto/ [obsoleto]

	MAX-IO	DEP-IO	*CODA/obstruent
a. o.so.le.to	*!		
b. $\varnothing$ ob.so.le.to			*
c. o.be.so.le.to		*!	

Despite the preservation of the obstruent, these dialects exhibit featural neutralizations in voice and continuancy (Hualde 1989a).¹⁰ In other words, the featural specification of the obstruent is altered with regard to the features [voice] and [continuant], incurring one or more faithfulness violations. The relevant faithfulness constraint is IDENT [feature] and its more specific versions IDENT(voice) and IDENT(continuant).

10. There is a significant amount of cross-dialectal variation regarding the [continuant] and [voice] specifications of coda obstruents that cannot be properly addressed within the confines of this paper. For instance, in the dialect of Northern and Northwestern Spain (colloquial, fast speech) described by Fernando Martínez Gil (2003) coda obstruents are always [+continuant], except for those cases in which an underlying voiceless stop is followed by a voiceless sop, e.g. *apto* [apto] 'apt'; but *hipnosis* [iβnósis] 'hypnosis'; *abdicar* [aβðikár] 'abdicate'; *absurdo* [aβsúrðo] 'absurd'. This suggests that AGREE (voice) (Voicing assimilation, in Martínez Gil [2003]), is highly ranked. Martínez Gil also argues for a highly ranked SPIR constraint that says that voiced stops are disallowed after continuant segments (i.e., voiced stops are realized as spirants). SPIR is therefore a more restricted version of AGREE (continuant). Although the goal here is not to propose an alternative analysis, I suspect that the spirantization effects can be obtained from more basic constraints, such as, for instance, a restriction on the combination of the features [–continuant] and [+voice]. AGREE (continuant) would be low ranked in this dialect, since voiceless stops are not disallowed before continuant segments, e.g., *excepción* [eksθepθion] 'exception'.

- (37) IDENT(feature): A segment's input specification for a specific feature must match that of the output.
 IDENT(voice): A segment's input specification for [voice] must match that of the output.
 IDENT(continuant): A segment's input specification for [continuant] must match that of the output.

For voicing neutralization, the output may reveal obstruent devoicing (/digno/ [dikno]) or assimilation (/futbol/ [fuðβol]). Since sonorants are not affected by devoicing, especially because voiceless sonorants are phonetically and typologically marked,¹¹ a markedness constraint against voiceless sonorants — *SONORANT [–voice] — must be undominated. However, for non-sonorants, the unmarked voice specification for the coda position is voiceless — *CODA [+voice] (voiceless codas are unmarked). *CODA [+voice] must dominate the relevant faithfulness constraint — IDENT(voice) —, so that it is possible to change input voice specifications, if necessary, to obtain a voiceless obstruent in the output. In sum, the relevant ranking for obstruent devoicing is:

- (38) Obstruent devoicing in coda position
 *SONORANT [–voice] >> *CODA [+voice] >> IDENT (voice)

- (39) Obstruent devoicing in coda position
 /digno/ [dikno]

	*CODA [+voice]	IDENT (voice)
a.  dik.no		*
b. dig.no	*!	

As seen in (39) the candidate with a voiced obstruent (39b), although faithful to the input, fails because it incurs a violation of the more highly ranked constraint against voiced obstruents in the coda.

Voice assimilation of coda obstruents responds to the domination of the constraint that requires adjacent segments to agree with respect to voice — AGREE (voice) — above IDENT (voice).

- (40) AGREE (voice): adjacent segments share the same specification for the feature (voice) (Lombardi 1999)
 AGREE (voice) >> IDENT (voice)

- (41) Regressive voice assimilation of coda obstruents
 /futbol/ [fuðβol]

	AGREE (voice)	IDENT(voice)
a.  fuðβol		*
b. futβol	*!	

(41a) is a better candidate than (41b) because it satisfies the dominating constraint AGREE [voice] by adopting the [+voice] specification of the following consonant.

11. Many languages that have voiced sonorants do not have voiceless ones. Those with voiceless sonorants, however, also have voiced ones.

Hualde (1989a) reports the presence of variation, even within speakers, between obstruent devoicing and voice assimilation, e.g. /futbol/ [fuðβol] ~ [futβol]. This suggests that the ranking of AGREE (voice) and *CODA [+voice] with respect to each other is not fixed. When AGREE (voice) and *CODA [+voice] are in conflict, the speaker must rank them, but, since the ranking is not fixed, sometimes AGREE (voice) will be ranked higher, whereas other times *CODA [+voice] will come out on top.

- (42) Different outcomes of voice neutralization in coda obstruents: Coda devoicing
/futbol/ [fuðβol]

	*CODA [+voice]	AGREE (voice)	IDENT (voice)
a. fuðβol	*!		*
b. $\varnothing$ futβol		*	

In (42) the output (42b) exhibits coda devoicing because *CODA [+voice] is more highly ranked than AGREE (voice). A violation of *CODA [+voice] eliminates the candidate with voice assimilation (42a).

- (43) Different outcomes of voice neutralization in coda obstruents: Voice assimilation
/futbol/ [futβol]

	AGREE (voice)	*CODA [+voice]	IDENT(voice)
a. $\varnothing$ fuðβol		*	*
b. futβol	*!		

(43) reveals the opposite situation to that in (42). The need to agree in voicing is more important than having a voiceless coda and therefore the candidate with assimilation, the one whose highest-ranked violation is *CODA [+voice], is the winner and the output (43a).

- (44) /fuDbol/ [futβol] (assuming [+ voice] in the input UR)¹²

	AGREE (voice)	*CODA [+voice]	IDENT(voice)
a. $\varnothing$ fuðβol		*	
b. futβol	*!		*

(44) is included to show that the results of evaluation would be the same regardless of the voice specification of the input. Voicing assimilation only affects coda obstruents, thus an obstruent in an onset cluster does not assimilate in voicing to the following sonorant, *apretar* [a.pre.tar] * [a.bre.tar] ‘to tighten’. This suggests that a more specific IDENT(voice) constraint — IDENT^{ONSET}(voice) — dominates AGREE (voice).

When the demands of AGREE (voice) and *CODA [+voice] are not in conflict, that is, when the onset consonant following the coda is voiceless and therefore voice assimilation would produce result in a voiceless consonant, there is rarely any alternation, as predicted: “*las sordas resultan prácticamente obligatorias si la consonante siguiente es una oclusiva sorda*” ‘voiceless segments are practically required if the following consonant is a voiceless stop’ (Hualde 1991, 33, translation mine)”. This is demonstrated in

12. D=voiced obstruent.

(45) and (46), where the output is invariantly [opsoleto], regardless of the ranking of AGREE (voice) and *CODA [+voice].

(45) /obsoleto/ [opsoleto]

	AGREE (voice)	*CODA [+voice]	IDENT(voice)
a. $\varnothing$ opsoleto			*
b. obsoleto, o β soleto	*!	*	

(46) /obsoleto/ [opsoleto]

	*CODA [+voice]	AGREE (voice)	IDENT(voice)
a. $\varnothing$ opsoleto			*
b. obsoleto, o β soleto	*!	*	

(38–46) focus on voicing, abstracting away from continuancy. Coda obstruents in this dialect of Spanish also show neutralization in continuancy, surfacing as either stops or fricatives. An OT explanation for the alternation lies in the fact that, by ranking markedness over faithfulness (IDENT(continuant)), preservation of a contrast is no longer important and the default, unmarked structure is preferred. On the basis of the sonority hierarchy (16), a coda fricative is less marked than a stop, /futbol/ [fuð β ol].

(47) Markedness >> IDENT(continuant)

*CODA/ stop >> *CODA/ fricative, IDENT(continuant)

However, in positions of prominence (stressed syllable, emphatic position, etc.) a stop is preferred for phonetic reasons. I use the cover term PROMINENCE [–cont] to refer to the default status of stops in prominent positions. PROMINENCE [–cont] dominates *CODA fricative, so that in a position of prominence a stop will be selected over a fricative, [fut β ol]

(48) PROMINENCE [–cont], *CODA/ stop >> *CODA/ fricative >> IDENT(continuant)

To recapitulate: an OT analysis makes the prediction that, under the general ranking Markedness >> Faithfulness, there will be various ways to satisfy *CODA/ obstruent, since there are various types of faithfulness constraints that could be violated (whole segment, featural identity, etc.). Such prediction is borne out by cross-dialectal Spanish data. In addition to the deletion of coda obstruents, and the voice and continuancy neutralizations seen here, other Spanish dialects avoid marked codas by incurring other faithfulness constraint violations: some, for instance, eliminate the supralaryngeal node of /s/ (MAX-IO(SL)), [meh] > /mes/ ‘month’; others vocalize obstruents (IDENT(Stricture)) (Piñeros 2001), /adkirir/ [a $\imath$ kirir] ‘acquire’. I will not present an OT analysis of all those processes because of space considerations. Such accounts go beyond the current goal of providing a comprehensive view of syllabification in Spanish in an OT framework highlighting OT’s advances. Obstruent coda deletion and voice and continuancy neutralizations are sufficient to show that an OT account can uncover the ‘hidden motivation’ behind all these processes (to eliminate a marked structure, a coda obstruent) in a way not possible under a serial analysis.

Before concluding the account of obstruents in coda position, it is important to bring to the reader's attention one last generalization with regard to the place features of coda obstruents (see Piñeros, this volume, for a complete account of place features in the coda). Unlike sonorants, which are usually coronal, obstruents in coda position retain their point of articulation, /fuDbol/ ~ [fuðβol] ~ [futβol] *[fupβol] *[fuββol]; /oBsoleto/ [opsoleto] ~ [oβsoleto] *[oðsoleto] ~ *[otsoleto]. This suggests the high-ranking of a constraint that requires preservation of place features in obstruents regardless of syllabic affiliation and of the well-known tendency of codas to neutralize place contrasts.

- (49) IDENT^{OBSTR}(place): The place features of an obstruent in the input must match that of the output.

CODA COND: A coda cannot license place features

IDENT^{OBSTR}(place) >> CODA COND

- (50) /obsoleto/ [opsoleto]

	IDENT ^{OBSTR} (place)	CODA COND	IDENT(place)
a. <i>o</i> opsoleto		*	
b. otsoleto	*!	*	*

Coda sonorants in Spanish tend to be realized as coronals (dorsals, in some dialects, cf. Piñeros, this volume, for an account that involves reducing consonantal features) or, in the case of nasals, to assimilate to the point of articulation of the following consonant. Hence it can be concluded that faithfulness to place features is not as important for sonorants as it is for obstruents. Ranking the more specific IDENT^{OBSTR}(place) over the general IDENT(place) will suffice to obtain this effect without having to propose an additional constraint that refers to sonorants (IDENT(place) is independently justified). However, despite the restrictions on point of articulation in the coda (CODA COND), segments need place features; therefore HAVE PLACE must dominate CODA COND.

- (51) HAVE PLACE: all segments must have place features

HAVE PLACE >> CODA COND

In the case of sonorants, given that IDENT(place) is low-ranked, HAVE PLACE will be satisfied by selecting the place features ranked lowest in the place hierarchy (the least marked features), i.e. coronal (52a).

- (52) a. Place Hierarchy: *DOR >> *LABIAL >> *CORONAL
b. HAVE PLACE >> *DOR >> *LABIAL >> *CORONAL >> IDENT(place)

Nasals and laterals take their place features from the following C through assimilation, thus satisfying CODA COND and HAVE PLACE simultaneously (place features are obtained and licensed through the onset) (see Piñeros, this volume, for a complete account).

(53) /tango/ [tango]

	IDENT ^{OBSTR} (place)	HAVE PLACE	CODA COND	IDENT(place)
a.  tango				*
b. tango			*!	
c. tango			*!	*

For ease of presentation an underlyingly specified coronal point of articulation is assumed in (53). This assumption has no bearing on the results of candidate evaluation; different assumptions about the nature of the underlying representation result in the selection of the same optimal candidate (53a) (Richness of the Base, McCarthy 2002).

1.2.2.1.2 Word-final codas. In word-final position only coronal consonants are possible: the sonorants /-l/, /n/, and /-r/ (*papel*, *camión*, *amor*) and the obstruents /-s/ and /θ/ (*mes*, *pez*), and /d/ (*virtud*).¹³ /d/ can be realized as [ð], [θ], [t] or be deleted. Other consonants appear in unassimilated borrowings, *club*, *frac*, *bulldog*, *album*, *chef* (Hualde 1999a). Except for the behavior of obstruents, these facts are generally in agreement with the account of word-medial codas in 1.2.2.1. Although word-medial stops and fricatives are acceptable in some dialects, as shown above, word-final obstruents are less common and even rare in some dialects.

Non-coronal word-final obstruents tend to appear in borrowings. The absence of word-final obstruents in the native lexicon is related to the fact that those segments were repaired at one point in the history of the language through epenthesis, so many of these forms are [e] final today. At the point that epenthesis was operative, the ranking would have been *CODA/ obstruent, MAX-IO >> DEP-IO. I argue, however, that the present ranking for many dialects, is *CODA/ obstruent, DEP-IO >> MAX-IO, as demonstrated by the outputs of foreign words, *club* [klú], *frac* [fra]. More specifically, since only non-coronal obstruents are affected (coronal obstruents are well-formed), the ranking must be:¹⁴

- (54) Non-coronal coda obstruents are deleted (rather altering place features). Coronal coda obstruents are retained
 *CODA/obstruent, DEP-IO >> MAX-IO
 IDENT^{OBSTR}(place), HAVE PLACE >> *DOR >> *LABIAL >> MAX-IO >> *CORONAL,
 IDENT(place)

That some varieties of Spanish preserve word-medial non-coronal coda obstruents can be explained by restricting the relevant ranking to Latinate words (nativized forms, part of the Spanish lexicon) in formal contexts.

13. Exceptionally, /-x/, *reloj*, and /-t/, *cénit*, can appear in final codas (Hualde 1999a, Harris 1983 disagrees)

14. Although deletion of word-final obstruents is a common phenomenon, some dialects exhibit retention and neutralization (in voice, and/or continuancy) as in word-medial position. Such dialects exhibit the rankings for word-medial obstruents in (35), (38), (47). Neutralization of place features would respond to the ranking: HAVE PLACE, MAX-IO, DEP-IO >> *DOR >> *LABIAL >> *CORONAL, IDENT(place), IDENT^{OBSTR}(place).

- (55) Preservation of the place features of word-internal non-coronal coda obstruents in
 latinate words (formal context)
 $\text{IDENT}^{\text{OBSTR}}(\text{place}), \text{HAVE PLACE}, \text{MAX-IO}, \text{DEP-IO} \gg *DOR \gg *LABIAL \gg$
 $*CORONAL, \text{IDENT}(\text{place})$ (latinate words)

1.2.2.1.3 Complex codas. Spanish has coda clusters and therefore $*\text{COMPLEX CODA}$ must be dominated by faithfulness constraints. Nonetheless, since not all coda clusters are possible, additional constraints must be involved. Most coda clusters contain a coda consonant or glide + /s/. /s/ is well known to have special status cross-linguistically that allows its adjunction to other segments, even in violation of sonority (*stop, abstract*). I will therefore not delve into this topic and will simply treat these clusters as exceptional. Codas consisting of a glide + /s/, *dais* ‘you give’, or a glide + coronal sonorant, *veinte* ‘twenty’, *aunque* ‘although’, conform to the sonority contour and to the coda markedness hierarchy that prefers coronal sonorants. Only /s/ is allowed after another consonant in a coda cluster (Hualde 1999a: 17), *biceps* ‘biceps’, *torax* ‘thorax’, *transporte* ‘transport’. In fast speech, these clusters are usually simplified to [bises], [toras], [trasporte], suggesting the ranking:

- (56) $*\text{COMPLEX CODA} \gg \text{MAX-IO}$

Yet (56) alone is not sufficient, as it does not say anything about which consonant in the cluster will be deleted. In an OT account, the universal sonority constraint hierarchy will select the correct output, deleting always the least sonorous consonant.

- (57) $*\text{COMPLEX CODA} \gg \text{MAX-IO}, *CODA/obstruent \gg *CODA/nasal \gg *CODA/liquid$
 $\gg *CODA/glide$

A serial analysis, however, faces a serious obstacle in trying to account for the same facts. It cannot satisfactorily explain the selection of the segment targeted for deletion. Harris (1983) argues that in *esculp-ir* ‘to sculpt’, *esculp-tor* ‘sculptor’ [p] is deleted in [escultor] because it cannot be syllabified in the coda or in the onset. Yet, this mechanism will not work for /ekspresidente/ [espresiðente] $*[ekpresiðente]$ because both [k] and [s] are syllabifiable in the coda (through R1 and R3 in Harris 1983). An additional deletion rule targeting [k] would be necessary to derive [espresiðente]. Similarly, in Hualde (1991) a rule of /s/-Adjunction (‘Adjoin /s/ under N’) must be followed by another rule that deletes a stop in a stop + /s/ cluster. In sum, to obtain the correct outputs, serial models must resort to adhoc rules that miss an important generalization — the target of deletion is the segment that makes the worst coda. In an OT account, this is a direct consequence of the universal sonority constraint hierarchy.

2. Syllabification rules. Resyllabification

In Section 1 I provided a general OT account of Spanish syllable types and phonotactics. I showed that the basic descriptive facts and syllabification mechanisms can be accounted for by means of the interaction of universal markedness and faithfulness constraints ranked in language/dialect-specific fashion. For comparison purposes it

is now pertinent to briefly review how a derivational approach would account for the same facts. Based on the ruled-based approach to syllabification of Steriade (1982) and the formalism of Levin (1985), Hualde (1991) proposes that syllabification in Spanish is the result of the application of the following set of ordered rules:

1. Node projection: Mark vowels as syllable-heads, create N nodes, and project N' and N'' nodes.
 2. Complex Nucleus: Adjoin a prevocalic glide under the N node.
 3. CV rule: Adjoin a consonant to the left of the nucleus under the N'' node.
 4. Complex onset: Adjoin a second consonant under the N'' node if the result would be a permissible onset cluster (stop or /f/ + liquid, except */dl/ and (*) /tl/).
 5. Coda rule: Adjoin a segment to the right of the nucleus under N'.
 6. Complex coda: Adjoin a consonant to the right of a glide under N'.
- /s/-Adjunction: Adjoin /s/ under N'.

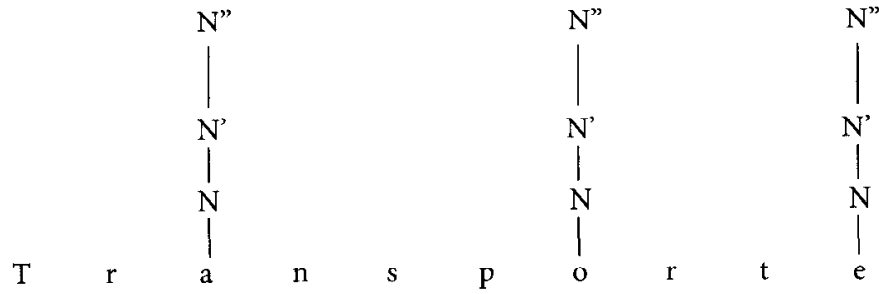
Figure 1. Spanish syllabification rules (Hualde 1991)

The CV rule has a second, structure-changing application. As mentioned above, an additional rule simplifying CsC clusters is also necessary (deletes a stop in a stop + /s/ cluster).

An account of syllabification like the one in *Figure 1* and *Figure 2* has the disadvantage of relying on language-specific rules. It also requires language-specific, extrinsic rule-ordering. Furthermore, although many of the generalizations expressed by the rules are common cross-linguistically, such as for instance, the fact that most languages seem to have an Onset Rule, this is only coincidental in a model of phonology where rules are language-specific. On the other hand, an OT analysis like the one presented in this paper can account for the cross-linguistic preferences present in the rules, in addition to replacing language specific rules with universal constraints. For instance, the preference for CV syllables is a consequence of a markedness constraint that penalizes onsetless syllables (or requires ONSET). Such constraint is phonetically grounded as it provides for a better sonority contour for the syllable. Similarly, *CODA is motivated by the universal dislike for codas and by the linguistic implicational that, although many languages do not allow codas, no language requires them. An additional inadequacy of the serial model is that the syllabification algorithm requires descriptive generalizations/conditions on rule application indicating what types of segments constitute well-formed onset, codas and nuclei, e.g., "4. Complex onset: Adjoin a second consonant under the N'' node *if the result would be a permissible onset cluster (stop or /f/ + liquid, except */dl/ and (*) /tl/).*" In OT all of this is obtained directly through the general mechanism of constraint interaction. Finally, one particular aspect of syllabification that reveals the superiority of an OT analysis is the process traditionally known as *resyllabification* — in Spanish a word final consonant is generally resyllabified as the onset of the following vowel-initial word.

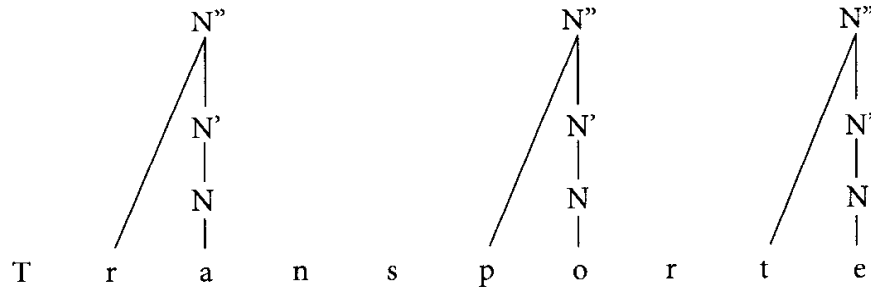
- | | | |
|------|---------------------------|---------------|
| (58) | los amigos [lo.sa.mi.yos] | 'the friends' |
| | más osos [má.so.sos] | 'more bears' |
| | reloj azul [re.lo.xa.sul] | 'blue watch' |

Rule 1

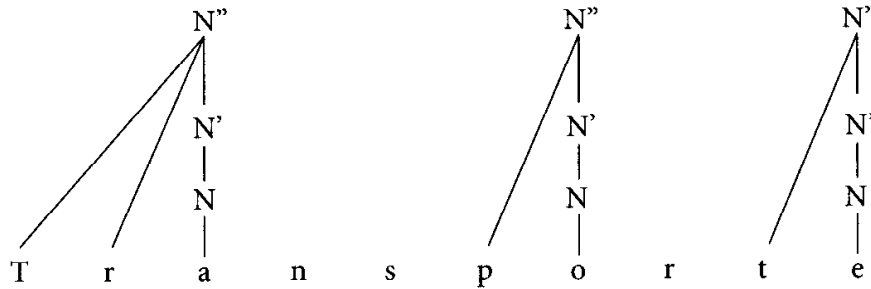


Rule 2 (N/A)

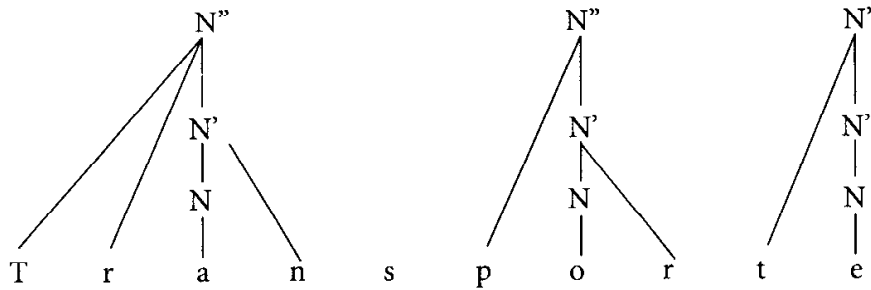
Rule 3



Rule 4



Rule 5



Rule 6 (N/A)

/s/-Adjunction

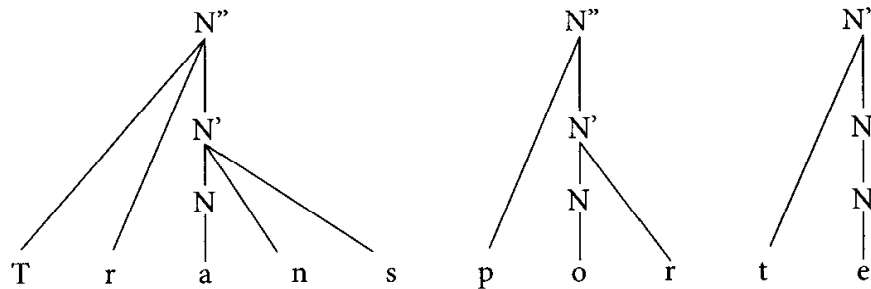


Figure 2. Application of Spanish syllabification rules (Hualde 1991)

In [kluβ.lin.do] ONSET satisfaction is no longer at stake. Therefore ALIGN-LST selects [kluβ.lin.do] over misaligned [klu.β|lin.do]. *CODA must be ranked below ALIGN-LST given that a coda consonant (*CODA violation) is preferred to misalignment (ALIGN-LST violation).

(66) ONSET >> ALIGN (Stem, L, Syllable, L) >> *CODA

(67) /klublindo/ [kluβ.lin.do]

	ONSET	ALIGN-LST	*CODA
a. $\mathcal{P}$ kluβ. lin.do			*
b. klu.β lin.do		*!	

Diphthogization applies across words in Spanish (i.e., syllable merger), *mi amigo* [mja.mi.ɣo] 'my friend' where it is also the result of the domination of ONSET over ALIGN-LST and MAX-IOμ.

(68) ONSET >> ALIGN-LST, MAX-IOμ.

(69) /miamigo/ [mja.mi.ɣo]

	ONSET	ALIGN-LST	MAX-IOμ
a. $\mathcal{P}$ mja.mi.ɬo		*	*
b. mi. a.mi.ɬo	*!		*

As seen in (69), (a) is the optimal candidate because (b) contains a violation of the top-ranked constraint, ONSET. (for a more in-depth OT analysis of syllable merger in Chicano Spanish see Bakovic, this volume, and Martínez-Gil 2004; other recent analysis of the phenomenon are Roca 1991, Hualde 1994, Martínez-Gil 2000).

3. Domain of syllabification. Interaction with morphology

Since rules apply in a certain domain, rule-based models of syllabification need to specify the domain of syllabification. On the basis of resyllabification, aspiration, and onset strengthening data, Hualde (1989b, 1991) convincingly argues that the domain of syllabification in Spanish is a unit smaller than the word, as syllabification seems to apply before the adjunction of prefixes and before compounding (after affixation).

In an OT analysis the issue of domain can be obtained directly from the general mechanisms, independently needed in the theory, that capture the interaction between morphology and syllabification/phonology. More specifically, this type of interaction is formalized through alignment constraints that refer to morphological and prosodic constituents (see Kager 1999:117–120 for the general format of alignment constraints). In Spanish the constraint responsible for word-level syllabification (before affixation and compounding) is ALIGN-LST (63).

As seen in (70), a candidate like (70b) [de.sje.lo] that avoids onset strengthening through resyllabification of the final consonant of the prefix as an onset is not possible because it would misalign the syllable and the stem (ALIGN-LST violation).

It is preferable to create an acceptable onset by underparsing a mora and changing the specification of the feature sonorant (MAX-IO μ and IDENT (son) violations). This indicates that ALIGN-LST dominates MAX-IO μ and IDENT (cont). It has already been shown that ONSET >> ALIGN-LST (64).

(70) /des+ ielo / [desyelo]

	ALIGN-LST
a. $\varnothing$ des. yelo	
b. de. s <u>ie</u> .lo	*!

(71) /des+ ielo/ [desyelo]

	*ONSET/glide	ONSET	ALIGN-LST	MAX-IO μ	IDENT (son)
a. $\varnothing$ des. yelo				*	*
b. de. s <u>ie</u> .lo			*!	*	
c. des. jelo	*!			*	

ALIGN-LST also selects the right candidate in forms like /sublunar/ [su β .lunar].

(72) /sublunar/ [su β .lunar]

	ONSET	ALIGN-LST	*CODA
a. $\varnothing$ su β . lu.nar			*
b. su. β lu.nar		*!	

That syllabification applies after suffixation shows the effects of ALIGN -R (73).

(73) ALIGN (Word, R, Syllable, R) (ALIGN -R): The right edge of a morphological word coincides with the right edge of a syllable.

Due to space limitations I will not present an account of the interaction between aspiration and syllabification in Spanish. The reader is referred to Hualde (1989b, 1991) for a derivational account and to Colina (2002) and Wiltshire (this volume) for optimality-theoretic approaches. I will, however, stress that in forms like *des-hacer* [de.ha.ser] 'to undo' the issue at stake is not one of alignment, given that prefixes are not morphological words¹⁶, but one of allomorph uniformity. In aspirating dialects of Spanish the final /s/ in /des/ will be realized as [h] when /des/ is attached to a consonant-initial stem, /des+kontrol/ [deh.kon.trol]. Some of these dialects also show aspiration when the prefix precedes a vowel initial stem, despite resyllabification of final /s/ to the onset position. The motivation behind this is the need to reduce allomorphy in /des/. The relevant constraint, IDENT (PrWd) (Colina 2002), requires identity amongst realizations of prosodic words. In some dialects, IDENT (PrWd) dominates faithfulness, and therefore the output contains aspiration, [de.ha.ser]. In others, however, the output is [de.sa.ser], which is the result of the opposite ranking IDENT -IO >> IDENT (PrWd) (see also Wiltshire, this volume, for an alternative account).

16. An alternative is to replace morphological word with prosodic word in the alignment constraint.

In sum, syllable structure and morphological structure (word-level) are usually isomorphic in Spanish, except when onset satisfaction is at stake, i.e. diphthongization, resyllabification. In OT terms, ALIGN-LST is a high-ranked constraint dominated by ONSET.

4. Syllable-repair mechanisms

In rule-based models of syllabification, repair of ill-formed syllables is done through additional rules of deletion or epenthesis (see for Spanish, Harris 1983, 1987, Hualde 1991). In some models, filters are also used to avoid the production of illicit outputs. In an OT approach, the traditional notions of epenthesis and deletion correspond to violations of faithfulness constraints (MAX-IO, DEP-IO, etc). Deletion and epenthesis are no longer the result of language-specific, often adhoc, rules; they can be obtained through the general mechanism of interaction of faithfulness and markedness constraints. *Syllable repair* is a syllabic consequence of the basic mechanism that drives the phonology — producing the best output possible.

Finally, one more advantage of an optimality theoretic model is that it can account for the quality of the epenthetic vowel, rather than stipulating it for each language (Spanish = [e]). Since epenthetic segments are by definition absent from the input, faithfulness constraints are trivially satisfied; as a result, markedness constraints become relevant. Hence it is not surprising that the epenthetic vowel should consist of the least marked vocalic features.

(74)		i	u	e	o	a
	high	+	+	–	–	–
	low	–	–	–	–	+
	back	–	+	–	+	–
	round	–	+	–	+	–

On the basis of the Spanish vowel inventory in (74), it is safe to assume that the marked, contrastive vowel features are [+low], [+round], [+high]. These are preserved when present in the input through the domination of the relevant faithfulness constraints over markedness:

- (75) IDENT (low), IDENT (round), IDENT (high), IDENT (back) >> *[+low], *[+round],
*[+high] *[+back]

When faithfulness constraints are trivially satisfied because no vocalic features are present in the input (epenthesis), markedness constraints emerge requiring an output that satisfies *[+low], *[+round], *[+high] *[+back]. Hence the output must be [–low], [–high], [–back], [–round] that is, [e], the default epenthetic vowel in Spanish.

5. Conclusions

In this article I provided an optimality-theoretic account of Spanish syllable structure that integrates all areas covered by traditional analyses of syllabification: syllable types, phonotactic restrictions, syllabification algorithms (or rules), domain of syllabification, and repair strategies. The descriptive facts in those areas can be explained through the general mechanism of interaction of universal markedness and faithfulness constraints. Cross-dialectal and intra-speaker variation is obtained through variation in the ranking of the constraints. This analysis also offers a comprehensive view of recent advances in Spanish syllabification that was well overdue, since the most recent studies of this type date back to the late 80s and early 90s (Harris 1989a, Hualde 1991). Yet the true driving force behind the writing of this article is the highlighting of the advantages of an optimality theoretic approach to syllabification in order to support the claim that the advent of OT has brought about significant improvement in syllabic theory. Among these advantages is the replacement of language-specific rules with universal constraints. A serial approach based on language-specific rules fails to capture/explain the similarity of syllabification rules across languages; since rules are independent, such similarity can only be the result of coincidence. That syllabification in OT is the result of the language-specific ranking of universal constraints provides an explanation for cross-linguistic similarities as well as variation. In addition, I show that serial analyses cannot capture the hidden motivation behind seemingly independent rules. In OT, such motivation is the result of trying to satisfy highly-ranked universal constraints through the violation of various other lower-ranked ones. For instance, the current analysis reveals that processes like onset strengthening, diphthongization, resyllabification all respond to the need to satisfy ONSET. An optimality-theoretic account also makes it possible to avoid the conditions on rule application (e.g. ‘Attach a second consonant to the onset, *if the resulting cluster is well-formed-i.e., obstruent or /f/ + liquid*’) that were necessary in rule-based models in order to achieve descriptive adequacy; they are now directly obtained through constraint interaction. Furthermore, an optimality-theoretic framework allows the analyst to put dialectal variation in a broader perspective: dialectal variation corresponds to various logical ways of solving a problem (of obtaining a more optimal output). The result is a more restrictive and explanatory view of variation than that offered by serial models, in which languages were different because their rules were different. Finally, generalizations which were stipulatory in derivational analyses are now predicted by the theory. That the epenthetic vowel in Spanish is /e/ is not an accident. As predicted, when faithfulness to the input is not a requirement — precisely the case for epenthetic segments which by definition have no input correspondent — the lower ranked markedness constraints emerge, demanding the least marked output [e].

References

- Baković, Eric. This volume. Hiatus resolution and cooperative interaction. In *Optimality-theoretic Studies in Spanish Phonology*, Fernando Martínez-Gil and Sonia Colina (eds). Amsterdam: John Benjamins.
- Barlow, J. and J. Gierut. 1999. Optimality theory in phonological acquisition. *Journal of Speech, Language, and Hearing Research* 42: 1482–98.
- Clements, N. 1990. The role of sonority in core syllabification. In *Papers in Laboratory Phonology I: Between the Grammar and Physics of Speech*, J. Kingston and M. Beckman (eds), 282–333. Cambridge: CUP.
- Colina, S. 1995. A Constraint-based Approach to Syllabification in Spanish, Catalan and Galician. PhD dissertation, University of Illinois, Urbana-Champaign.
- Colina, S. 1996. Spanish truncation processes: The emergence of the unmarked. *Linguistics* 34: 1199–1218.
- Colina, S. 1997. Identity constraints and Spanish resyllabification. *Lingua* 103: 1–23.
- Colina, S. 1999. Reexamining Spanish glides: Analogically conditioned variation in vocoid sequences in Spanish dialects. In *Advances in Hispanic Linguistics. Papers from the 2nd Hispanic Linguistics Symposium*, J. Gutiérrez-Rexach and F. Martínez-Gil (eds), Vol. 1, 121–134. Somerville MA: Cascadilla.
- Colina, S. 2002. An account of inter and intradialectal variation in Spanish /s/ aspiration. In *Structure Meaning and Acquisition in Spanish. Papers from the 4th Hispanic Linguistics Symposium*, J. Lee, K. Geeslin and J. Clancy Clements (eds), 230–243. Somerville MA: Cascadilla.
- Gnanadesikan, A. 2004. Markedness and faithfulness constraints in child phonology. In *Constraints in Phonological Acquisition*, R. Kager, J. Pater and K. Zonneveld (eds), 73–108. Cambridge: CUP.
- Harris, J.W. 1983. *Syllable Structure and Stress in Spanish*. Cambridge MA: The MIT Press.
- Harris, J.W. 1987. Epenthesis processes in Spanish. In *Studies in Romance Languages*, C. Neidle and R. Núñez-Cedeño (eds), 107–122. Dordrecht: Foris.
- Harris, J.W. 1989a. Our present understanding of Spanish syllable structure. In *American Spanish Pronunciation*, P.C. Bjarkman and R.M. Hammond (eds), 151–169. Washington DC: Georgetown University Press.
- Harris, J.W. 1989b. Sonority and syllabification in Spanish. In *Selected Papers from the Linguistic Symposium on Romance Languages* 17, C. Kirschner and J. DeCesaris (eds), 139–153. Amsterdam: John Benjamins.
- Hooper, J. 1976. *An Introduction to Natural Generative Phonology*. New York NY: Academic Press.
- Kahn, D. 1976. *Syllable-based Generalizations in English Phonology*. Cambridge MA: The MIT Press.
- Hualde, J.I. 1989a. Procesos consonánticos y estructuras geométricas en español. *Lingüística* 1: 7–44.
- Hualde, J.I. 1989b. Silabeo y estructura morféica en español. *Hispania* 72: 821–831.
- Hualde, J.I. 1991. On Spanish syllabification. In *Current Studies in Spanish Linguistics*, H. Campos and F. Martínez-Gil (eds), 475–493. Washington DC: Georgetown University Press.
- Hualde, J.I. 1994. La contracción silábica en español. In *Gramática del español*, V. Demonte (ed.), 629–647. México DF: El Colegio de México.
- Hualde, J.I. 1997. Spanish /i/ and related sounds: an exercise in phonemic analysis. *Studies in the Linguistic Sciences* 27: 61–79.
- Hualde, J.I. 1999a. La silabificación en español. In *Fonología de la lengua española contemporánea*, R. Núñez-Cedeño and A. Morales-Front (eds), Washington DC: Georgetown University Press.
- Hualde, J.I. 1999b. Patterns in the lexicon: Hiatus with high unstressed high vowels in Spanish. In *Advances in Hispanic Linguistics. Papers from the 2nd Hispanic Linguistics Symposium*, J. Gutiérrez-Rexach and F. Martínez-Gil (eds), Vol. 1, 182–197. Somerville MA: Cascadilla.

- Hualde, J.I and Prieto, M. 2002. On the diphthong/hiatu contrast in Spanish: Some experimental results. *Linguistics* 40: 217–234.
- Kager, R. 1999. *Optimality Theory*. Cambridge: CUP.
- Kenstowicz, M. and Kisseberth, C. 1979. *Generative Phonology*. New York NY: Academic Press.
- Itô, J. and A. Mester. 1994. Reflections on CodaCond and alignment. In *Phonology at Santa Cruz III*, J. Merchant, J. Padgett and R. Walker (eds), 27–46. Santa Cruz CA: LRC.
- Levin, J. 1985. *A Metrical Theory of Syllabicity*. PhD dissertation, MIT.
- Lombardi, L. 1999. Positional faithfulness and voicing assimilation in optimality theory. *Natural Language and Linguistic Theory* 17: 267–302.
- Martínez-Gil, F. 1996. El principio de la *distancia mínima de sonoridad* y el problema de la vocalización consonántica en el español dialectal de Chile. *Hispanic Linguistics* 8: 201–246.
- Martínez-Gil, F. 1997. Obstruent vocalization in Chilean Spanish: A serial versus a constraint-based approach. *Probus* 9: 165–200.
- Martínez-Gil, F. 2000. La estructura prosódica y la especificación vocálica en español: El problema de la sinalefa en ciertas variedades de la lengua coloquial contemporánea. In *Panorama de la fonología española actual*, J. Gil Fernández, (ed.), 511–560. Madrid: Arco Libros, S.L.
- Martínez-Gil, F. 2001. Sonority as a primitive phonological feature. In *Features and Interfaces in Romance*, J. Herschensohn, E. Mallén and K. Zagana (eds), 203–222. Amsterdam: John Benjamins.
- Martínez-Gil, F. 2003. Resolving ordering paradoxes of serial derivations: An optimality-theoretical account of the interaction of spirantization and voicing assimilation in Peninsular Spanish. In *Theory, Practice, and Acquisition: Papers from the 6th Hispanic Linguistics Symposium and the 5th Conference on the Acquisition of Spanish and Portuguese*, P. Kempchinsky and C.E. Piñeros (eds), 40–67. Somerville MA: Cascadilla.
- Martínez-Gil, F. 2004. Hiatus resolution in Chicano Spanish. Paper presented at the 34th Linguistics Symposium on Romance Languages, University of Utah. Salt Lake City UT.
- McCarthy, J. 2002. *A Thematic Guide to Optimality Theory*. Cambridge: CUP.
- McCarthy, J. and A. Prince. 1993a. *Prosodic Morphology I: Constraint Interaction and Satisfaction*. Report no. RuCCS-TR-3. New Brunswick NJ: Rutgers University Center for Cognitive Science.
- McCarthy, J. and A. Prince. 1993b. Generalized alignment. In *Yearbook of Morphology*, G. Booij and J. van Marle (eds), 79–153. Dordrecht: Kluwer.
- McCarthy, J. and A. Prince. 1994. The emergence of the unmarked. In *Proceedings of the North East Linguistic Society* 24, M. González (ed), 333–79. Amherst MA: GLSA.
- Morales-Front, A. 1994. *A Constraint-based Approach to Spanish Phonology*. PhD dissertation, University of Illinois, Urbana-Champaign.
- Núñez-Cedeño, R.A. and Morales-Front, A. 1999. *Fonología generativa contemporánea de la lengua española*. Washington DC: Georgetown University Press.
- Ohala, D. 1999. The influence of sonority on children's cluster reductions. *Journal of Communication Disorders* 32: 397–422.
- Piñeros, C.E. 2000. Prosodic and segmental unmarkedness in Spanish truncation. *Linguistics* 38: 63–98.
- Piñeros, C.E. 2001. Segment-to segment alignment and vocalization in Chilean Spanish. *Lingua* 111: 163–188.
- Piñeros, C.E. This volume. The phonology of implosive nasals in five Spanish dialects: An optimality account. In *Optimality-theoretic Studies in Spanish Phonology*, F. Martínez-Gil and S. Colina (eds). Amsterdam: John Benjamins.
- Prince, A. and P. Smolensky. 1993. *Optimality Theory: Constraint Interaction in Generative Grammar*. Report no. RuCCS-TR-2. New Brunswick NJ: Rutgers University Center for Cognitive Science.
- Rice, K. 1992. On deriving sonority: A structural account of sonority relationships. *Phonology* 9: 61–100.

- Roca, I. 1991. Stress and syllables in Spanish. In *Current Studies in Spanish Linguistics*, H. Campos and F. Martínez-Gil (eds), 599–635. Washington DC: Georgetown University Press.
- Selkirk, E.O. 1984. On the major class features and syllable theory. In *Language Sound Structure*, M. Aronoff and R.T. Oehrle (eds), 107–136. Cambridge MA: The MIT Press.
- Steriade, D. 1982. Greek Prosodies and the Nature of Syllabification. PhD dissertation, MIT.
- Wiltshire, C. This volume. Prefix boundaries in Spanish varieties: A non-derivational OT account. In *Optimality-theoretic Studies in Spanish phonology*, F. Martínez-Gil and S. Colina (eds). Amsterdam: John Benjamins.

Appendix 1. List of constraints and constraint ranking by process

Segment types in the nucleus

*NUC/obstruent >> *NUC/nasal >> *NUC/liquid >> FAITH, *NUC/vowel

Segment types in the onset

*ONSET/vowel >> FAITH >> *ONSET/ liquid >> *ONSET/ nasal >> *ONSET/obstruent

Segment types in the coda

*CODA/vowel>>FAITH>>*CODA/obstruent>>*CODA/nasal>>*CODA/liquid>> *CODA/glide

Diphthongization: [piérðe] > /pierce/

*ONSET/glide, ONSET >> *COMPLEX NUCLEUS, *NUC/glide, MAX-IO_μ

ONSET >> *COMPLEX NUCLEUS, *CODA/glide, *CODA, MAX-IO_μ

Onset strengthening: [kre.yeŋ.do] > /kre-iendo/

*ONSET/glide, ONSET >> MAX-IO_μ, IDENT (son)

Coda deletion: [osoletto] > /obsoleto/

*CODA/obstruent, DEP-IO >> MAX-IO

Coda retention: [obsoleto] > /obsoleto/

DEP-IO, MAX-IO >> *CODA/obstruent

Obstruent devoicing in coda position: [dikno] > /digno/

* SONORANT [–voice] >> *CODA [+voice] >> AGREE (voice)>> IDENT (voice)

Regressive voice assimilation of coda obstruents: [fuðβol] > /futbol/

AGREE (voice)>> *CODA [+voice]>> IDENT (voice)

Continuacy neutralization: [fuðβol] ~ [futβol] > /futbol/

*CODA stop>> *CODA fricative, IDENT(continuant), PROMINENCE [–cont],*CODA stop >> *CODA fricative >> IDENT(continuant)

Place features retained in obstruents: [fuðβol] ~ [futβol] *[fupβol] *[fuββol] > /fuDbol/

IDENT^{OBSTR}(place) >> IDENT(place)

IDENT^{OBSTR}(place)>> CODA COND

Non-coronal coda obstruents deleted. [klú] >/klub/

*CODA obstruent, DEP-IO >> MAX-IO

Coronal coda obstruents retained [mes] > /mes/

IDENT^{OBSTR}(place), HAVE PLACE >> *DOR >> *LABIAL >> MAX-IO>> *CORONAL, IDENT(place)

Preservation of place features of word-internal non-coronal coda obstruents in latinate words (formal context) [dikno] > /digno/

IDENT^{OBSTR}(place), HAVE PLACE, MAX-IO, DEP-IO >> *DOR >> *LABIAL >> *CORONAL, IDENT(place) (latinate words)

Complex codas [bises] < /bisepts/

*COMPLEX CODA >> MAX-IO

*COMPLEX CODA >> MAX-IO, *CODA/obstruent >> *CODA /nasal >> *CODA /liquid >> *CODA /glide

Resyllabification [má.so.sos] [kluβ.lin.do]

ONSET >> ALIGN (Stem, L, Syllable, L) >> *CODA

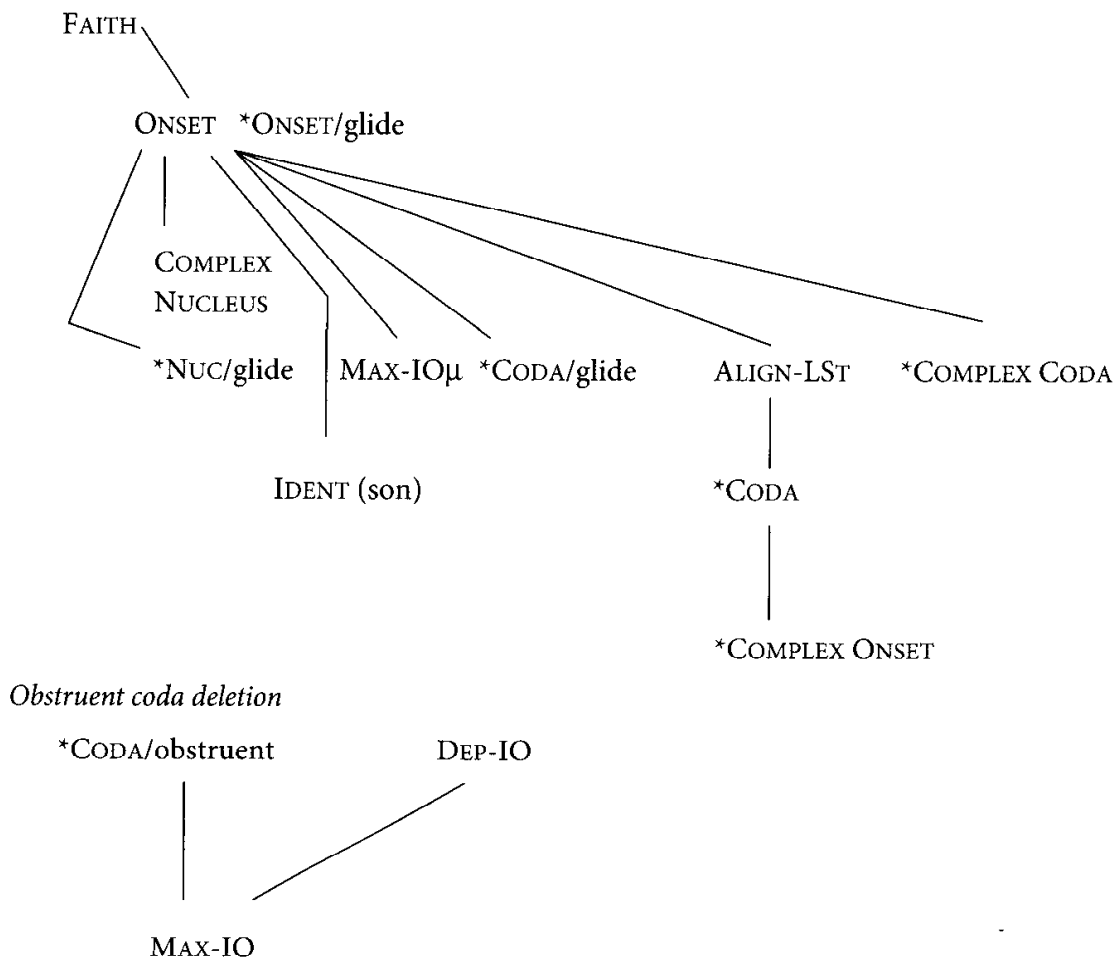
Diphthongization across words [mja.mi.yo] > /miamigo/

ONSET >> ALIGN-LST, MAX-IO_μ

Onset strengthening across words [desyelo] > /des+ ielo /

ONSET /glide, ONSET >> ALIGN-LST >> MAX-IO_μ, IDENT (son)

Appendix 2. Summary of constraint rankings



Preservation of the place features of word-internal non-coronal coda obstruents in **latinate** words (formal context)

IDENT^{OBSTR}(place), HAVE PLACE

|

MAX-IO, DEP-IO

|

*DOR

|

*LABIAL

|

*CORONAL, IDENT(place)

Complex codas

*COMPLEX CODA

|

MAX-IO, *CODA/obstruent

|

*Coda/nasal

|

*Coda/liquid

|

*CODA/glide